Series Intro: Java 25 Design Patterns

Season: 0 · Intro · Category: Overview · Level: Beginner

Intent

- Map the journey from GoF classics through enterprise/concurrency to pure FP with Zazr and VAVR.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- A gym curriculum: fundamentals, compounds, then advanced Olympic lifts.

When to Use

- Starting interview prep or modernizing a Java codebase.
- The design force named in Intent shows up more than once.

When Not to Use

- Skipping straight to Free monads before understanding Strategy.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```
// see README.md + pom.xml (Java 25, Zazr 0.4.0, VAVR 1.1.0)
```

Modern Java 25 / Functional

- Java 25: records, sealed interfaces, pattern matching, virtual threads. Zazr is a Vavr fork built for those features.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: README.md + pom.xml (Java 25, Zazr 0.4.0, VAVR 1.1.0)
```

Pitfalls

- Don't memorize UML — understand the force that the pattern resolves.
- Prefer composition and sealed types over deep inheritance.

Interview Q&A

- Q: Why learn patterns in 2026? A: They name recurring design forces; interviews and code reviews still use that vocabulary. Modern Java m...
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Map the journey from GoF classics through enterprise/concurrency to pure FP with Zazr and VAVR.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: A gym curriculum: fundamentals, compounds, then advanced Olympic lifts. [0:40] Intent — State the de...